Sustainability

Energy Consumption and Optimization Strategies of Cloud-Based Big Data and Machine Learning Applications: Current Trends and Future Directions:

- Cloud data centers account for 1-2% of the global electricity consumption
- A **multi-layered approach** (algorithmic improvements, infrastructure, and renewable energy) is crucial for balancing computational needs and environmental responsibility

Why Transparency matters for sustainable data centers and carbon-neutral artificial intelligence (AI)

- Data centers could consume 9% of global electricity demand by 2030
- Lack of transparency in data sharing leads to researchers facing limitations in developing effective solutions to minimize carbon emissions
- Divides proposal/methodology in three sections for a total of a 10 year plan

Implementations

Forecasting power-efficiency related key performance indicators for modern data centers using LSTMs

- Introduces ML-based (- forecasting framework) approaches aiming to mode and evaluate various data center energy/power consumption-related KPIs
- The energy consumption of the cooling infrastructure approximately accounts for 40% of the total data center power consumption.

Adaptive, Efficient, and Fair Resource Allocation in Cloud Datacenters leveraging Weighted A3C Deep Reinforcement Learning (2025):

- **Problem Statement:** Cloud data centres demand adaptive, efficient, and fair resource allocation techniques due to heterogeneous workloads with varying priorities. However, most existing approaches struggle to cope with dynamic traffic patterns, often resulting in suboptimal fairness, increased latency, and higher energy consumption.
- Unlike static rule-based schedulers, WA3C continuously learns from the environment, making it resilient to changing workload patterns and system dynamics.
- **Result**: WA3C achieves more effective convergence, lower latency, and balanced resource allocation among jobs

Empowering energy sustainability: A cutting-edge ensemble network for precise power forecasting (2025)

- **Purpose**: forecasting model for electrical energy consumption is demanded by power management systems
- **Problem**: Mainstream energy forecasting methods (using ML and DL methods) contain non-linearity between input and output → requires more improvement robustness, performance and generalisation.
- **Proposed solution:** a hybrid forecasting model with CNN and GRU and Self attention mechanism
    - CNN: extracts spatial information in historical input data
    - GRU: ephasizes temporal information sequentially

- SA: extracted features are fed into SA module to acquire dominant patterns for final forecasting
- **Results**: CNNGRU-SA model is compared with competitive techniques over power generation and consumption benchmarks. Achieves higher performance and outperforms other models (Metrics: MSE, MAE and RMSE)
- The proposed framework enhances the development of intelligent energy systems supporting more effective and efficient power management within smart grid infrastructures

Reinforcement learning for data center energy efficiency optimization (2025)

- RL and DRL demonstrated potential in improving data center energy efficiency
- **Problem statement:** a comprehensive review of the deployment of these algorithms remain limited
- **Result**: systeati review - exploring the application of RL/DRL algorithms for optimizing daa center energy efficiency - focusing on optimizing the operation of cooling systems and ICT processes:
    - **Task scheduling:** the process of mapping user tasks to available virtual machines (VMs) or resources to optimize performance, minimize latency, and reduce costs
    - **Resource allocation:** the dynamic process of distributing, managing, and provisioning virtualized resources—such as CPU, memory, storage, and network bandwidth—from data centers to user applications
    - **VM consolidation/placement**
    - **Network traffic control**
- Benchmark comparisons, experimental setup, datasets and implementation platforms
- Highlights what is missing: research, lack of real-time validation for developed algorithms, absence of multi-scale standardized metrics for reporting energy efficiency improvements

DynamoLLM *(2024)*

Provided Literature:

On the diversity of cluster workloads and its impact on research results *(2018)*

- Aim: Proved overfitting of Google's dataset and released four analyzed traces → "largest cluster with a publicly available trace"
- Google released a set of scheduler logs that is used in 450 publications - leads to researchers to overfit their work to google's dataset characteristics
- Prove the importance of dataset plurality and diversity → by evaluating the performance of a job runtime predictor
- **Terminology**:
    - Cluster Workload (and Private Clustered Workload)
    - High Performance Computing (HPC) clusters
    - Job Runtime Predictor
    - Scheduler logs

Failures in Large Scale Systems: Long-term measurement, Analysis and Implications (2017)

- **Goal**: Compare and contrast the **reliability** characteristics of multiple large-scale HPC production systems
- **Methodology**: covers more than 1 billion compute node hours across 5 different systems over a period of 8 years

CarbonScaler: Leveraging Cloud Workload Elasticity for Optimizing Carbon-Efficiency (2023)

- Cloud platforms want to reduce their operational carbon footprint
- A common approach is to exploit the temporal flexibility by executing them in periods with the greenest energy and suspending them at other times.
- Suspend-resume approaches incur long delays in **job completion times**
- **Goal:** Implementation of CarbonScaler - uses autoscaling capabilities to guide carbon-efficient deployment of batch applications in the cloud
- Notion of carbon scaling - exploits the elasticity of batch workloads in the cloud to optimize their carbon emissions.

GreenOps

Cloud Computing Datasets
https://github.com/ACAT-SCUT/Awesome-CloudComputing-Datasets

Hugging face AI energy benchmark:
https://huggingface.github.io/AIEnergyScore/#related-work
https://huggingface.co/spaces/AIEnergyScore/Leaderboard

Ideas:
Impact on water